Check for updates

View Article Online
View Journal

PCCP
Physical Chemistry Chemical Physics

Accepted Manuscript

This article can be cited before page numbers have been issued, to do this please use: Y. Xie, M. Li and C. Li, *Phys. Chem. Chem. Phys.*, 2026, DOI: 10.1039/D5CP05056A.



This is an Accepted Manuscript, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. We will replace this Accepted Manuscript with the edited and formatted Advance Article as soon as it is available.

You can find more information about Accepted Manuscripts in the Information for Authors.

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard Terms & Conditions and the Ethical guidelines still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this Accepted Manuscript or any consequences arising from the use of any information it contains.

ROYAL SOCIETY OF CHEMISTRY

rsc.li/pccp

View Article Online
DOI: 10.1039/D5CP05056A

Pressure-Induced Phase Transitions in Co_3O_4: a First-principles Study

Yaping Xie[1,2], Ming Li[2,*], Cong Li[1,*]

[1]*Key Laboratory of Intelligent Optoelectronic Devices and Chips of Jiangsu Higher Education Institutions, School of Physical Science and Technology, Suzhou University of Science and Technology, Suzhou, 215009, China*

[2]*School of Chemical Engineering, North China University of Science and Technology, Tangshan, 063210, China*

*Corresponding Author Email: vrchem@hotmail.com; cong.li@hpstar.ac.cn

Abstract: Co_3O_4 is an important transition-metal oxide with wide applications in catalysis and energy storage. In this work, using evolutionary *ab initio* structural searches combined with first-principles calculations, we systematically investigated its structural stability, phase transitions, magnetic evolution and electronic properties within the pressure range of 0-100 GPa. Our results show that the cubic *Fd-3m* phase first transforms into an orthorhombic *Fddd* structure through a symmetry-lowering distortion driven by compressibility mismatch between $Co^{2+}O_4$ and $Co^{3+}O_6$ polyhedra. Pressure-enhanced Co–O hybridization and broadened 3*d* bandwidth modify the exchange interactions, driving the magnetic ground state from antiferromagnetic (*Fd-3m*) to ferromagnetic (*Fddd*). A subsequent first-order transition to a monoclinic $P2_1/c$ phase involves full CoO_6 coordination, lattice densification, and charge redistribution between Co^{2+} and Co^{3+}, ultimately leading to magnetic collapse. Based on these results, a detailed temperature–pressure phase diagram is constructed. These findings not only elucidate the intrinsic coupling among crystal structure, coordination environment, and magnetism in Co_3O_4 under high pressure, but also provide insight into phase transitions in spinel-type transition-metal oxides.

Keywords: Transition Metal Oxides; High Pressure; Phase Transition; Magnetism

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

1. Introduction

Spinel-type oxides with the general formula AB_2O_4 are widely used in materials science and energy-related applications. In this class of materials, cobalt (Co) can occupy either the A or B crystallographic site, while the other metal site is typically occupied by a transition metal or a rare-earth element. These oxides exhibit a wide range of physicochemical properties. Among them, spinel cobalt oxide Co_3O_4 has attracted considerable research interest in recent years due to its p-type semiconducting nature and the presence of highly active crystal facets, particularly in the fields of catalysis, energy storage, and electronic devices[1-5].

Under ambient conditions, Co_3O_4 crystallizes in a cubic spinel structure with space group *Fd-3m* (No. 227). In this structure, cobalt ions exist in two distinct oxidation states: divalent Co^{2+} ions occupy the tetrahedral sites and adopt a high-spin configuration, whereas trivalent Co^{3+} ions reside at the octahedral sites in a low-spin state. This spin-state differentiation gives rise to overall antiferromagnetic ordering and leads to pronounced spin-lattice coupling effects in spinel Co_3O_4.[6] Due to its unique structure and magnetic properties, Co_3O_4 has been extensively studied under ambient conditions, including solid-state sensors[7,8], electrochemical devices[9], lithium batteries [4,10,11] and hydrogen production[12]. It has also gained attention as a promising material in emerging technologies, such as catalysis for low-temperature CO oxidation and the oxygen evolution reaction[13,14].

Given the well-established behavior of Co_3O_4 at ambient conditions, the application of external pressure offers an intriguing avenue to further tune its structural and magnetic properties. High pressure can shorten interatomic distances, enhance orbital overlap, and potentially induce structural phase transitions or magnetic transformations, thereby providing a new route for engineering its properties. Indeed, magnetic metals such as Fe[15] and Co[16], covalent solids such as Si[17] and P[18], and molecular solids like H_2[19,20] exhibit rich high-pressure phase behavior, making them prototypical representatives of metallic, covalent, and molecular systems, respectively.

Motivated by this general sensitivity of materials to compression, several studies

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

have explored the high-pressure behavior of Co_3O_4. Early studies[21,22] suggested that the cubic spinel structure of Co_3O_4 can be retained up to 42 GPa, accompanied by a pressure-induced charge-transfer transition from the normal to the inverse spinel at 17.7 GPa. Subsequently, pressure-induced structural transformations in the range of 30–46 GPa were reported[23], leading to the formation of a monoclinic $P2_1/c$ phase, which was attributed to progressive charge redistribution between Co^{2+} and Co^{3+} ions; upon decompression below 28 GPa, another monoclinic phase with space group *C2/m* was proposed. Later first-principles study[24,25] questioned the stability of the *C2/m* phase and suggested that a local high-spin to low-spin transition of Co^{2+} ions may precede the *Fd-3m* to $P2_1/c$ structural transformation. Recently, high-pressure X-ray diffraction (XRD) and Raman spectroscopy measurements[26] revealed a more complex, multi-step structural evolution under compression, involving successive transitions from the cubic phase to an orthorhombic phase at ~23 GPa, followed by monoclinic phases at ~45 and ~52 GPa. These pressure-induced changes were further supported by XANES and Raman spectroscopy observations[27]. Despite these efforts, notable discrepancies remain regarding the sequence of phase transitions and, in particular, the detailed atomic configurations of the high-pressure phases, underscoring the need for further systematic investigation.

Several first-principles studies have previously examined the high-pressure behavior of Co_3O_4. For example, Kaewmaraya *et al.*[24] reported a first-order transition from the cubic *Fd-3m* phase to a monoclinic $P2_1/c$ structure at ~35 GPa, associated with a high-spin to low-spin transition and charge transfer between Co^{2+} and Co^{3+}. Sousa *et al.*[25] investigated pressures up to 30 GPa and proposed a pressure-induced magnetic collapse within the spinel framework without global symmetry breaking. However, the relative stability of competing magnetic configurations over an extended pressure range, as well as the detailed coupling between structural and magnetic evolution, remains incompletely understood. In particular, the proposed transition sequences and magnetic behaviors are not yet fully consistent across theoretical and experimental reports.

In this work, we employ global structural search combined with first-principles calculations to systematically investigate Co_3O_4 over 0–100 GPa. By quantifying lattice

View Article Online
DOI: 10.1039/D5CP05056A

distortion, coordination reconstruction, and electronic structure evolution, we establish a unified structural–magnetic phase diagram and clarify the microscopic mechanism of the pressure-driven transitions. Our results show that pressure-induced modification of Co–O bonding progressively alters exchange interactions, driving the antiferromagnetic-to-ferromagnetic transition across the *Fd-3*m → *Fddd* transformation and ultimately leading to magnetic collapse in the high-pressure $P2_1/c$ phase. This study provides a coherent picture of the coupled lattice, electronic, and magnetic evolution of Co_3O_4 under compression.

2. Computational details

To systematically search for thermodynamically stable structural candidates of Co_3O_4 under high pressure, structure prediction was performed using the CALYPSO code based on the particle swarm optimization (PSO) algorithm[28,29], in combination with first-principles calculations implemented in the Vienna ab initio Simulation Package (VASP)[30,31]. Several compounds[32-35] were successfully predicted by CALYPSO code. Within the framework of Density Functional Theory (DFT), the Perdew-Burke-Ernzerhof (PBE) generalized gradient approximation (GGA)[36] was employed, and structural optimizations as well as electronic structure calculations were carried out using VASP[30,31]. Considering the strong correlation effects of Co-3*d* electrons, the DFT+U method[37] was used to calculate the electronic and magnetic properties of Co3O4 (see details in Section 3.1). The electron-ion interactions were described by the projector augmented-wave (PAW) method[38]. The valence electronic configurations were taken as $3p^6 3d^8 4s^1$ for Co atoms and $2s^2 2p^4$ for O atoms. To ensure the convergence of enthalpy calculations, a plane-wave cutoff energy of 800 eV was adopted, together with appropriate Monkhorst-Pack k-point meshes[39]. Phonon dispersion relations were calculated using the finite-displacement method as implemented in the PHONOPY package[40, 41], with a 3×3×3 supercell constructed to ensure the accuracy of all predicted stable structures. To further improve the precision of atomic force calculations, the total energy convergence criterion was set to 10^{-8} eV,

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

and a k-point grid density of $0.03\times2\pi$ Å^{-1} was used. The calculations of the electronic charge transfer were based on the Bader charge analysis[42, 43], which is an intuitive way of studying charge distribution the charge.

3. Results and discussion

3.1 Benchmarking DFT+U descriptions

Under ambient conditions, Co_3O_4 is a correlated transition-metal oxide in which the localized Co–3*d* electrons strongly influence its magnetic and electronic properties. It is well known that standard DFT calculations tend to underestimate band gaps in such systems due to self-interaction effects. More advanced many-body approaches, such as dynamical mean-field theory (DMFT)[44], can provide a more rigorous treatment of strong correlation effects. However, the primary objective of the present work is not to quantitatively resolve many-body correlation phenomena, but rather to establish a self-consistent microscopic understanding of pressure-induced structural and magnetic transitions. For this purpose, we adopt the DFT+U approach, which effectively incorporates on-site Coulomb interactions and improves the description of localized 3*d* states.[45, 46] To assess the sensitivity of different physical properties to on-site electron correlations, we benchmark the conventional DFT approach against DFT+U calculations for the ambient-pressure cubic *Fd-3m* phase of Co_3O_4.

The calculated lattice parameter, magnetic moment, and band gaps of *Fd-3m* phase as functions of the Hubbard U value are summarized in Fig. 1. By systematically comparing the DFT+U results with available experimental data[21, 23], we find that the calculated lattice parameter (Fig. 1(a)) progressively deviates from the experimental value with increasing U, whereas the best agreement is obtained within standard DFT (U = 0 eV). This indicates that the equilibrium structural properties of Co_3O_4 are relatively insensitive to on-site Coulomb corrections within the considered U range. In contrast, both the magnetic moment (Fig. 1(b)) and band gaps (Fig. 1(c)) are significantly underestimated in standard DFT calculations. As U value increases, the magnetic moment increases and approaches the experimental value[47], showing a

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
https://doi.org/10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

saturation behavior around U = 3.0 eV. Meanwhile, the calculated band gap exhibits an almost linear dependence on U and reaches good agreement with experiment[48-50] at U = 3.0 eV.

A direct comparison with the DFT+U results reported by Cadi-Essadek *et al.*[51] is also presented in Fig. 1. Our calculated magnetic moments and band gaps are in very good agreement with their results over the same U range. However, a noticeable deviation is observed in the lattice parameter. This difference originates primarily from the choice of pseudopotential. In the present work, the Co_pv pseudopotential includes the $3p^6 3d^8 4s^1$ electrons in the valence, whereas Cadi-Essadek *et al.*[51] employed a Co pseudopotential without explicit 3*p* states in the valence. To clarify this effect, we performed additional calculations using the Co pseudopotential ($3d^8 4s^1$ valence), and the resulting lattice parameters closely reproduce those of Ref. 51, as shown in Fig. 1(a). This comparison indicates that the inclusion of Co 3*p* semicore states moderately affects the equilibrium lattice parameter. Notably, when U = 0 eV, the Co_pv pseudopotential yields lattice parameters in closer agreement with experiment[21, 23].

Based on these benchmarks, standard DFT is adopted for structural optimization to ensure accurate lattice parameters, while DFT+U with U = 3.0 eV is employed for the calculation of magnetic and electronic properties in the following sections. We note that a U value in the range of 2.5–3.5 eV has been widely used in previous theoretical studies[51-53] of Co_3O_4 and related systems[54, 55], providing a physically reasonable description of the localized Co 3*d* states.

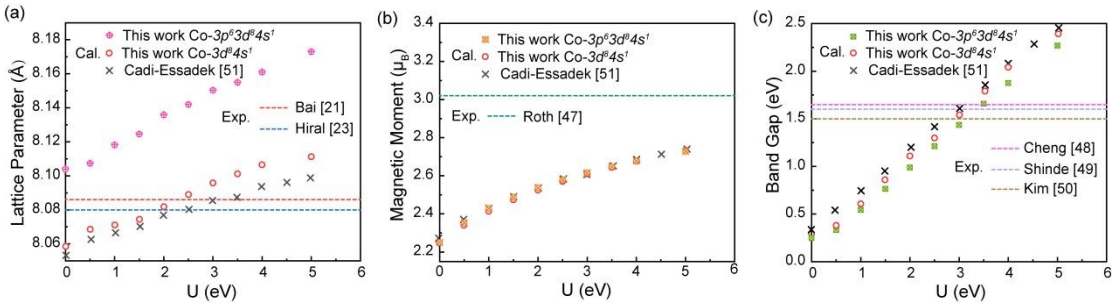


Fig. 1 Lattice parameter, local magnetic moment of Co^{2+}, and electronic band gap as a function of the Hubbard U parameter for the ambient-pressure cubic *Fd-3m* phase of Co_3O_4. The dashed lines represent the corresponding experimental values[21, 23, 47-50]. Open cross symbols denote the present

View Article Online
DOI: 10.1039/D5CP05056A

calculations using the Co_pv pseudopotential ($3p^6 3d^8 4s^1$), while open red circles correspond to the calculations performed with the Co pseudopotential ($3d^8 4s^1$). Solid black cross symbols represent the DFT+U results reported by Cadi-Essadek *et al.*[51], obtained using the $3d^8 4s^1$ valence configuration of Co.

3.2 Phase transitions and structural properties

We performed extensive structure searches for Co_3O_4 at pressures of 0 GPa, 50 GPa, and 100 GPa, with simulation cells containing up to 4 formula units. In total, 6 candidate structures were obtained over the pressure range of 0-100 GPa, including the *Fd-3m*, *Fddd*, *P2$_1$/c*, *C2/m*, *Cmcm* and *Pbcm* phases, among which the cubic *Fd-3m* structure corresponds to the stable spinel phase under ambient conditions. After further rigorous geometry optimizations within standard DFT, the pressure-dependent enthalpy differences of these candidate phases relative to the ambient-pressure *Fd-3m* phase are shown in Fig. 2(a). With increasing pressure, the enthalpy difference between the cubic *Fd-3m* phase and the orthorhombic *Fddd* phase gradually decreased and approached zero at approximately 30 GPa. Upon further compression to about 44 GPa, the monoclinic *P2$_1$/c* phase became energetically favorable, as its enthalpy reached the lowest value among all competing phases, signaling the occurrence of a first-order phase transition. The crystallographic parameters of the identified high-pressure phases are summarized in Table 1. To assess the influence of on-site electron correlations on the structural phase stability, we additionally calculated the enthalpy evolution of the competing phases using the DFT+U (U = 3.0 eV), as shown in Fig. 2(b). The overall phase transition sequence obtained from DFT+U remains identical to that predicted by standard DFT, although the corresponding transition pressures show moderate shifts: the *Fd-3m* to *Fddd* transition occurs at 24.7 GPa, followed by a transition to the *P2$_1$/c* phase at 42.7 GPa. Notably, the transition pressures obtained from standard DFT are closer to the experimentally reported XRD values[26, 27]. These results indicate that while the inclusion of the Hubbard U term slightly modifies the energetic balance between competing phases, the structural phase transition sequence itself is robust and weakly sensitive to on-site correlation effects.

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

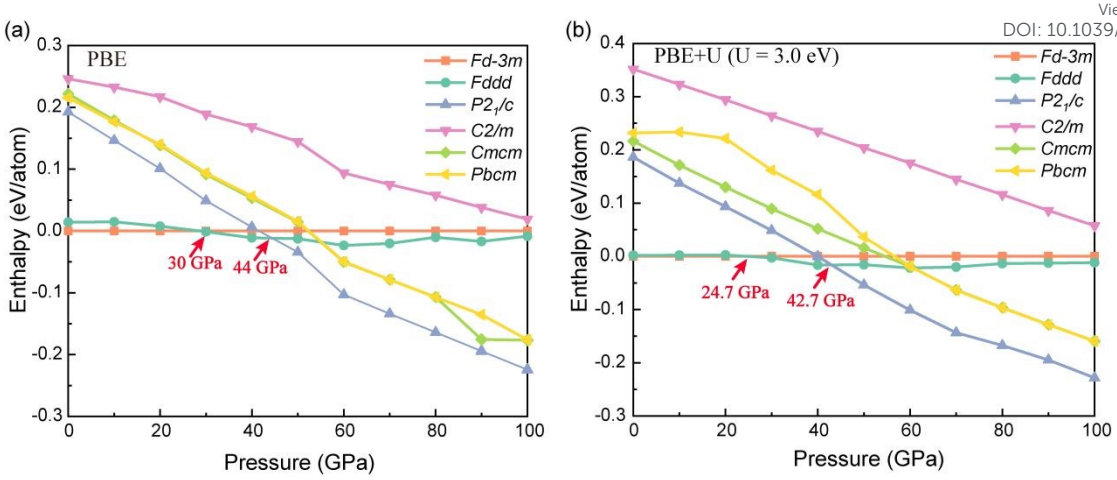


Fig. 2 Pressure-dependent relative enthalpies of competing Co_3O_4 structures with respect to the cubic *Fd-3m* structure, obtained from (a) standard PBE calculations and (b) PBE+U calculations with U = 3.0 eV.

Table 1. Calculated crystallographic parameters and atomic coordinates of Co_3O_4 in the *Fd-3m*, *Fddd*, and *P2₁/c* phases, together with the bulk modulus (B_0) and its first pressure derivative (B_0') obtained from fitting to the fourth-order Birch-Murnaghan equation of state.

Phase	Pressure (GPa)	Lattice Parameters (Å, °)	Atoms	Wyckoff Positions (fractional)			B_0 (GPa)	B_0'
				x	*y*	*z*		
Fd-3m	0 GPa	$a = b = c = 8.104$ $\alpha = \beta = \gamma = 90.000$	Co1 (8a)	0.000	0.000	0.000	201	4.6
			Co2 (16d)	0.625	0.625	0.625		
			O (32e)	0.000	0.500	0.500		
Fddd	30 GPa	$a = 7.825$ $b = 7.816$ $c = 7.807$ $\alpha = \beta = \gamma = 90.000$	Co1 (8b)	0.125	0.125	0.125	215	4.8
			Co2 (16c)	0.000	0.000	0.000		
			O (8a)	0.125	0.125	0.625		
P2₁/c	50 GPa	$a = 2.695$ $b = 8.786$ $c = 4.657$ $\alpha = \gamma = 90.000$ $\beta = 108.995$	Co1 (2b)	0.500	0.000	0.000	205	4.3
			Co2 (4e)	0.260	0.852	0.473		
			O1 (4e)	0.867	-0.047	0.714		
			O2 (4e)	0.570	0.685	0.691		

Based on the structure searches and enthalpy analyses, three energetically stable phases of Co_3O_4 were identified, namely the cubic *Fd-3m*, orthorhombic *Fddd* and monoclinic *P2₁/c*. Their corresponding crystal structures are illustrated in Figs. 3(a)-3(c). At ambient pressure, Co^{2+} (Co(II)) ions of cubic spinel *Fd-3m* structure occupy the 8a sites (0, 0, 0) and are coordinated by four O atoms to form CoO_4 tetrahedra,

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

whereas Co^{3+} (Co(III)) ions resided at the 16d sites (0.625, 0.625, 0.625) and are octahedrally coordinated by six O atoms, forming CoO_6 units. The calculated (Fig. 3(d)). This indicates that the tetrahedrally coordinated Co^{2+} units are more compressible than the octahedral Co^{3+} units. This unequal compressibility introduces internal strain in the cubic spinel framework, as the shared oxygen atoms cannot simultaneously accommodate uniform contraction of both coordination polyhedra. To release the accumulated elastic strain, the system lowers its symmetry from cubic *Fd-3m* to orthorhombic *Fddd*, allowing anisotropic lattice distortion and subtle rearrangement of oxygen positions. As shown in Fig. 3(e), the *Fddd* structure exhibits slight anisotropy in the lattice parameters ($a \neq b \neq c$), reflecting pressure-induced distortion of the original cubic lattice. The transition occurs at approximately 30 GPa and is characterized by a continuous lattice distortion without a pronounced volume collapse (Fig. 3(f)), indicating a second-order–like structural transition. Importantly, the *Fddd* phase largely preserves the polyhedral topology of the cubic structure, retaining $Co^{2+}O_4$ tetrahedra and $Co^{3+}O_6$ octahedra.

Upon further compression to about 44 GPa, the orthorhombic *Fddd* phase becomes unstable and transforms into a monoclinic $P2_1/c$ structure through a discontinuous first-order transition, accompanied by an abrupt ~9.6% volume reduction (Fig. 3(f)). This substantial volume collapse signals a reconstructive transformation rather than a simple symmetry lowering. The most striking structural change is the elimination of tetrahedral Co^{2+} coordination. In the $P2_1/c$ phase, all Co ions adopt octahedral environments, leading to a fully sixfold-coordinated framework. This coordination reorganization fundamentally alters the topology of the cation network. Instead of the spinel-type arrangement of interconnected tetrahedra and octahedra, the structure evolves into a denser octahedral framework characterized by anisotropic stacking along the [001] direction. This reconfiguration significantly reduces the distinction between Co^{2+} and Co^{3+} sublattices. For example, at 45 GPa, the Co^{2+}–Co^{2+} and Co^{3+}–Co^{3+} separations both converge to approximately 2.71 Å, in contrast to the more differentiated network in the cubic phase. Such homogenization of cation separations indicates enhanced direct cation–cation interactions under high compression. Moreover, both CoO_6 polyhedra

View Article Online
DOI: 10.1039/D5CP05056A

exhibit noticeable distortion, reflecting strong anisotropic lattice compression and maximized packing efficiency. Our identified $P2_1/c$ structure consistent with that reported by Kaewmaraya *et al.* [24], further supporting the reliability of the monoclinic phase.

By fitting the volume-energy data using the fourth-order Birch-Murnaghan (B-M) equation of state, the bulk moduli (B_0) and their first pressure derivatives (B_0') were obtained for each phase. As summarized in Table 1, the bulk modulus of the $P2_1/c$ phase was 205 GPa with a B_0' value of 4.3, which was comparable to that of the cubic *Fd-3m* phase (B_0 = 201 GPa and B_0' = 4.6), indicating that the two phases possess similar mechanical properties.

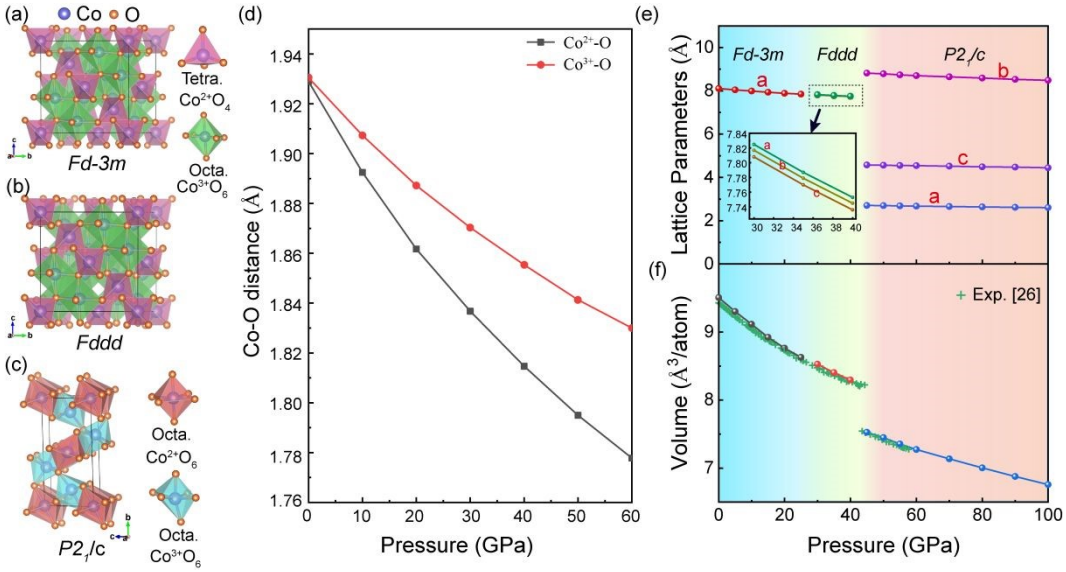


Fig. 3 (a–c) Crystal structures of Co_3O_4 in the *Fd-3m*, *Fddd*, and $P2_1/c$ phases, respectively. The dark red and green polyhedra represent the $Co^{2+}O_4$ tetrahedra and $Co^{3+}O_6$ octahedra in the *Fd-3m* and *Fddd* structures, respectively. In the $P2_1/c$ phase, both Co^{2+} and Co^{3+} ions are sixfold coordinated, forming $Co^{2+}O_6$ and $Co^{3+}O_6$ octahedra. (d) Pressure dependence of Co-O bond lengths of *Fd-3m*. (e) Pressure dependence of the lattice parameters for the three phases; the inset shows an enlarged view of the lattice-parameter evolution in the *Fddd* phase. (f) Pressure dependence of the unit-cell volume for the three phases compared with available experimental data[26]. Notably, the *Fddd* → $P2_1/c$ phase transition is accompanied by a volume collapse of approximately 9.6%.

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Although high-pressure XRD experiments have determined the lattice parameters of Co_3O_4 under compression, the precise atomic positions have not yet been resolved. In particular, the atomic configuration of the high-pressure monoclinic phases has often been assumed by analogy with the isostructural compound Ag_2O rather than being directly refined. Therefore, an independent structural validation is necessary to confirm the microscopic arrangement. We further calculated simulated XRD patterns for the *Fd-3m*, *Fddd*, and *P2₁/c* phases based on the fully relaxed atomic structures and compared them with available experimental data[26], as shown in Fig. 4. The calculated diffraction peak positions and relative intensities were in excellent agreement with the experimental data at the respective pressures[26], supporting the pressure-induced structural evolution of Co_3O_4. It is worth noting that another monoclinic phase (*C2/m*) was experimentally observed during decompression in the pressure range of 28 GPa[26]. However, our calculations indicate that the enthalpy of this phase is significantly higher than those of the *Fd-3m* and *P2₁/c* phases, indicating that this phase is not thermodynamically favored under compression at 0 K, while its appearance during decompression may be related to kinetic or metastable trapping.

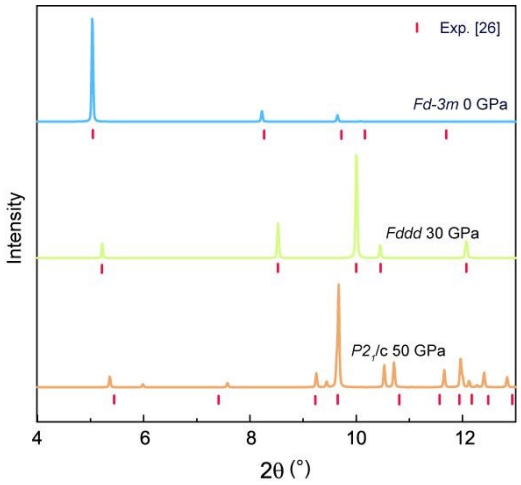


Fig. 4 Comparison between the simulated XRD patterns of the three phases and the experimental results[26].

3.3 Phonon spectra and dynamical stability

To evaluate the dynamical stability of the high-pressure phases of Co_3O_4, phonon spectra were calculated for each phase using the finite-displacement method within

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

supercells. Figures 5(a)-5(c) shows the phonon dispersion relations and the corresponding phonon density of states (PDOS) for the *Fd-3m* phase at 0 GPa, the *Fddd* phase at 30 GPa, and the $P2_1/c$ phase at 50 GPa, respectively. Within their respective stability pressure ranges, no imaginary phonon modes are observed, indicating that all three phases are dynamically stable under the corresponding pressures. For the ambient-pressure *Fd-3m* structure, the low-frequency region mainly originates from vibrations of Co^{2+} ions, whereas the high frequency region is dominated by vibrations associated with Co^{3+}-O bonds, reflecting the stronger bonding character of Co^{3+}. When the pressure is up to 30 GPa, the cubic structure undergoes a distortion and transforms into the orthorhombic *Fddd* phase. The phonon dispersion and PDOS of the *Fddd* phase remain largely similar to those of the *Fd-3m* phase, which reflects the similarity in local coordination environments and polyhedral topology between the two structures. However, when further compression induces the transition from the *Fddd* phase to the $P2_1/c$ phase, the phonon spectra exhibit pronounced reconstruction. A softening tendency emerges in the low- to intermediate-frequency regions, arising from the strong vibrational coupling between Co^{2+} and Co^{3+} ions. Nevertheless, no imaginary frequencies appear throughout the entire phonon dispersion, indicating that the $P2_1/c$ phase is dynamically stable within the investigated pressure range. At this stage, the phonon frequencies are clearly separated into two energy regions: the low to intermediate frequency vibrations are almost entirely contributed by the cooperative motions of Co^{2+} and Co^{3+} ions, indicating that the interactions between neighboring Co cations are significantly enhanced under pressure. This behavior is closely correlated with the discontinuous volume change observed during the first-order *Fddd* to $P2_1/c$ phase transition. In contrast, the high-frequency vibrational modes shift markedly toward higher energies, which can be attributed to the pronounced shortening of Co-O bond lengths and result in stronger bonding interactions.

To further assess the dynamical stability of the *C2/m* phase, we calculated its phonon spectrum at 50 GPa. As shown in Fig. 5(d), pronounced imaginary frequencies appear in the acoustic branches near the high-symmetry Y point, indicating lattice instabilities associated with long-wavelength distortions.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

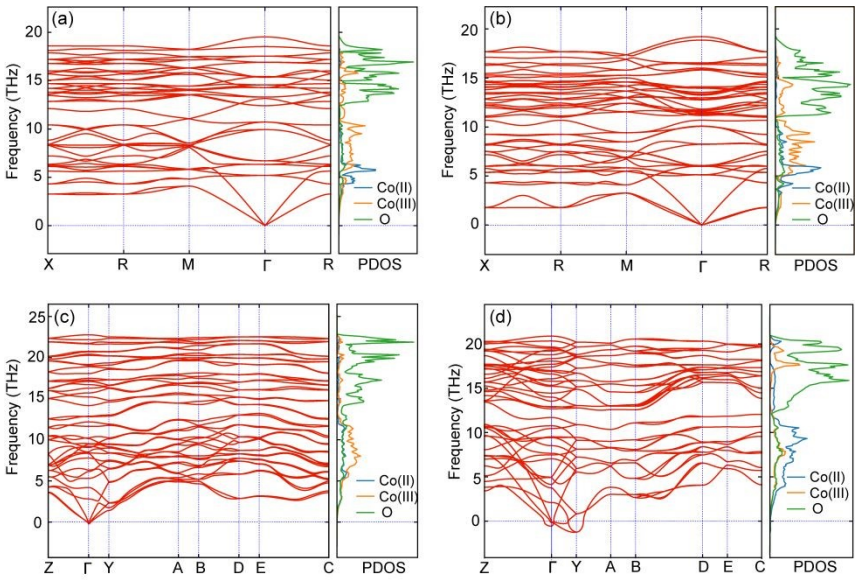


Fig. 5 Phonon dispersions and corresponding phonon density of states of Co_3O_4 in different phases: (a) cubic *Fd-3m* phase at 0 GPa; (b) orthorhombic *Fddd* phase at 30 GPa; (c) monoclinic $P2_1/c$ phase at 50 GPa.; (d) monoclinic *C2/m* structure at 50 GPa. Imaginary frequencies appear in the *C2/m* phase, indicating its dynamical instability under high pressure.

3.4 Magnetic properties

To identify the magnetic ground state of the stable phases, we have explicitly performed total-energy comparisons among antiferromagnetic (AFM), ferromagnetic (FM), and nonmagnetic (NM) configurations for the *Fd-3m* and *Fddd* phases within the 0-100 GPa range. The calculated total energies, summarized in Fig. 6(a), indicate that the AFM configuration is energetically favored for the *Fd-3m* phase, consistent with previous theoretical reports[24, 25], whereas the FM configuration has the lowest energy for the *Fddd* phase. Analysis of the local magnetic moments reveals that magnetism mainly originates from the Co^{2+} ions, while Co^{3+} ions carry negligible magnetic moments in both the *Fd-3m* and *Fddd* phases.

To further quantify the magnetic interactions, we evaluated the nearest-neighbor exchange coupling constant J_1 between Co^{2+} ions using the energy difference between AFM and FM configurations: $J_1 = \frac{1}{2} \times \frac{1}{4} \times \frac{1}{S^2} \times (E_{AFM} - E_{FM})$, where S = 3/2 for Co^{2+}, and E_{AFM} and E_{FM} are the total energies of unit cell with the AFM and FM states, respectively. The calculated value of J_1 = -1.21 meV for the *Fd-3m* at 0 GPa, is close

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

to the value reported in a previous experimental investigation[56]. For the *Fddd* phase at 30 GPa, J_1 is found to be 2.41 meV. These results confirm that the *Fd-3m* and *Fddd* phases favor AFM and FM ground states, respectively. For the high-pressure monoclinic $P2_1/c$ phase, the local magnetic moments collapse to zero after structural relaxation, regardless of the initial magnetic configuration (AFM or FM). Calculations initialized with different spin arrangements consistently converge to the same nonmagnetic solution, indicating that the nonmagnetic state represents the intrinsic ground state of the $P2_1/c$ phase under compression.

The evolution of magnetic moment under compression using the PBE+U method is shown in Fig. 6(b). At ambient pressure, the cubic spinel *Fd-3m* structure exhibit an AFM ground state, in which the tetrahedrally coordinated Co^{2+} ions carry a local magnetic moment of approximately 2.62 μ_B, whereas the octahedrally coordinated Co^{3+} ions remain in a low-spin state and contribute negligibly to the total magnetization. When the pressure reaches 30 GPa, the system undergoes a structural transition from the *Fd-3m* phase to the *Fddd* phase, accompanied by a magnetic transition from an antiferromagnetic to a ferromagnetic state. In the *Fddd* phase, the previously antiparallel Co^{2+} magnetic moments are reoriented and become ferromagnetically coupled, with a magnetic moment of about 2.55 μ_B. Upon further compression, the *Fddd* phase transforms into the monoclinic $P2_1/c$ structure, at which point the local magnetic moments of both Co^{2+} and Co^{3+} ions collapse simultaneously and the system enters a nonmagnetic state. This magnetic collapse indicates a substantial reorganization of the electronic states under high pressure, resulting in complete spin pairing and a nonmagnetic ground state.

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

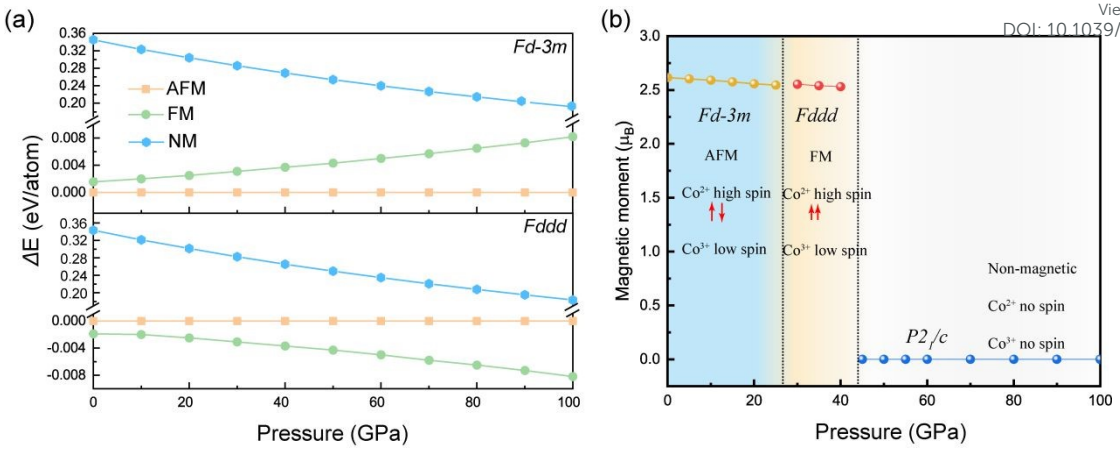


Fig. 6 (a) Total-energy comparison of the *Fd-3m* and *Fddd* phases in the AFM, FM, and NM configurations. (b) Pressure-induced magnetic transitions and evolution of magnetic moments in Co_3O_4. With increasing pressure, the system undergoes successive transitions from an AFM *Fd-3m* phase to a FM *Fddd* phase, and finally to a NM $P2_1/c$ phase, accompanied by a gradual reduction and eventual collapse of the magnetic moment.

3.5 Electronic Properties

To gain further insight into the electronic origin of the pressure-induced structural and magnetic transitions in Co_3O_4, we calculated the band structures and projected density of states (PDOS) for the representative phases at different pressures using DFT+U, as shown in Fig. 7.

At ambient pressure, the cubic *Fd-3m* phase exhibits a semiconducting character with a calculated band gap of 1.42 eV (Fig. 7(a)). This value falls within the range of the most frequently reported experimental optical gaps (~1.4–1.6 eV) and is in very good agreement with previous DFT+U results[51] obtained using comparable U parameters. A recent comprehensive review[57] systematically analyzed the optical band gap problem of bulk Co_3O_4 and clarified that three distinct experimental features have been reported: a low-energy band at ~0.78 eV, an intermediate band around ~1.51 eV, and a higher-energy transition near ~2.14 eV. Using an embedded-cluster approach combined with advanced multi-reference wavefunction-based methods (CASSCF/NEVPT2), the authors explicitly treated strong electron correlation effects and disentangled the contributions from high-spin tetrahedral Co^{2+} and low-spin

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

octahedral Co^{3+} centers. Their results demonstrate that the lowest feature corresponds to ligand-field excitations within Co^{2+} centers, the intermediate band involves mixed ligand-field and metal-to-metal charge transfer (MMCT) character, and the highest-energy band contains significant ligand-to-metal charge transfer (LMCT) contributions. Within the DFT+U framework, our calculated Kohn–Sham gap of 1.42 eV corresponds closely to the intermediate optical band (~1.5 eV), which is widely regarded as the fundamental semiconducting gap of bulk Co_3O_4. The lower-energy excitation (~0.78 eV) arises from localized ligand-field transitions of multi-reference character and therefore lies beyond the scope of single-determinant DFT+U. Likewise, the higher-energy feature (~2.14 eV) involves substantial charge-transfer contributions (LMCT/MMCT) and represents an excited-state optical transition rather than the fundamental band gap defined by the valence-to-conduction band separation in the Kohn–Sham picture.

The PDOS further confirms the electronic origin of this band gap. The valence band maximum (VBM) is primarily dominated by Co^{3+}-3*d* t_{2g} states noticeable hybridized with O-2*p* orbitals, while the e_g states remain largely unoccupied and form the conduction band minimum (CBM). The nearly symmetric spin-resolved DOS of Co^{3+} confirms a low-spin configuration $t_{2g}^6 e_g^0$ ($S = 0$), consistent with octahedral crystal-field splitting. In contrast, the tetrahedrally coordinated Co^{2+} exhibits pronounced spin asymmetry. The spin-up states of the t_{2g} orbitals are located in the valence band, while the spin-down states are located in the conduction band. The calculated DOS shows clear exchange splitting of these states, with the majority-spin components mainly located in the valence band and the minority-spin components shifted toward higher energies. This unequal occupation of spin channels corresponds to a high-spin configuration $e_g^4 t_{2g}^3$ ($S = 3/2$), indicating that the magnetic moment of *Fd-3m* Co_3O_4 originates from the partially filled Co^{2+}-3*d* states. Overall, orbital-resolved DOS of cubic phase obtained in the present work closely reproduce the electronic characteristics reported in the previous theoretical work[24, 25, 51], despite the use of a different pseudopotential treatment for Co. The minor quantitative differences

View Article Online
DOI: 10.1039/D5CP05056A

in band gap and bandwidth do not alter the underlying physical picture of crystal-field splitting and spin-state configuration in Co_3O_4.

Upon compression to the orthorhombic *Fddd* phase, the system retains semiconducting behavior with bandgap of approximately ~1.47 eV (Fig. 7(b)). However, noticeable modifications in the electronic structure are observed. Compared with the *Fd-3m* phase, the Co-3*d* bandwidth becomes broader, indicating enhanced Co–O hybridization under pressure. For octahedrally coordinated Co^{3+} ions, the spin-resolved DOS remains nearly symmetric, preserving the low-spin $t_{2g}^6 e_g^0$ configuration. Nevertheless, the t_{2g} manifold exhibits slight broadening and increased overlap with O-2*p* states, reflecting strengthened covalency and enhanced crystal-field splitting under compression. The tetrahedrally coordinated Co^{2+} ions retain pronounced spin asymmetry in their projected DOS. The Td crystal-field splitting into lower-energy e_g and higher-energy t_{2g} states is still evident, and exchange splitting remains finite, confirming that the high-spin configuration (FM) persists in this phase. The change in magnetic ordering from AFM to FM can therefore be attributed not to a collapse of the local high-spin state of Co^{2+}, but rather to a reconfiguration of exchange interactions driven by modifications in Co–O–Co bond geometry and enhanced *d–p–d* hybridization.

A qualitatively different electronic regime emerges in the high-pressure $P2_1/c$ phase. Although the system remains semiconducting, the band gap is substantially reduced to approximately ~0.86 eV, accompanied by a pronounced broadening of the Co-3*d* states. A key structural feature of this phase is the disappearance of tetrahedral sites, leading to octahedral coordination for all Co ions. As a consequence, the crystal-field environment is fundamentally altered. The enhanced octahedral crystal-field splitting and increased Co–O covalency result in substantial modification of the 3*d* electronic structure. The spin-resolved DOS becomes nearly symmetric for all Co ions, indicating the collapse of exchange splitting. Concurrently, the redistribution of spectral weight and the reduced separation between t_{2g} and e_g states suggest that the increased bandwidth under pressure suppresses the local magnetic moments. Furthermore, the shortened Co^{2+}–Co^{3+} distances in the $P2_1/c$ phase facilitate charge transfer between the

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

two cobalt species. As shown in Fig. 8, the Bader charges of Co^{2+} and Co^{3+} become nearly identical in this phase, in contrast to the more differentiated charge distribution in the *Fd-3m* and *Fddd* structures, consistent with previous electronic-structure analyses of monoclinic Co_3O_4 reported by Kaewmaraya *et al.*[24] These results indicate that the magnetic collapse in the $P2_1/c$ phase originates from coordination-driven electronic reorganization and intersite charge redistribution, rather than a simple compression effect. The continuous reduction of the band gap with pressure (Fig. 7(d)) further supports this picture and implies that continued compression may eventually drive the system toward metallization.

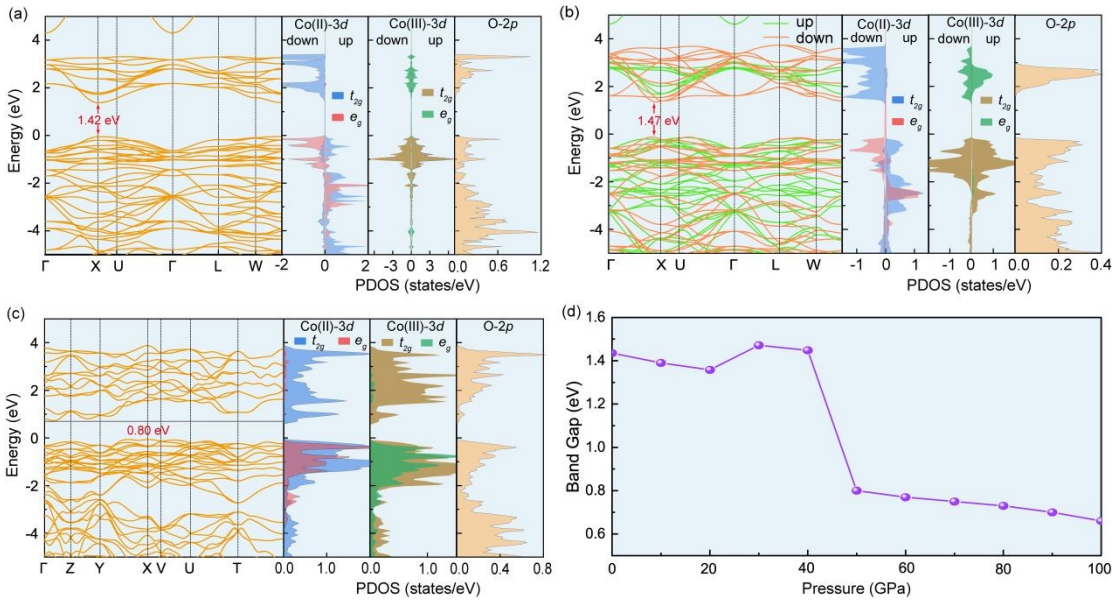


Fig. 7 Calculated band structures and projected density of states (PDOS) of Co_3O_4 using DFT+U with U = 3.0 eV, (a) the *Fd-3m* phase at 0 GPa, (b) the *Fddd* phase at 30 GPa, and (c) the $P2_1/c$ phase at 50 GPa. (d) Pressure-dependent band gap.

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

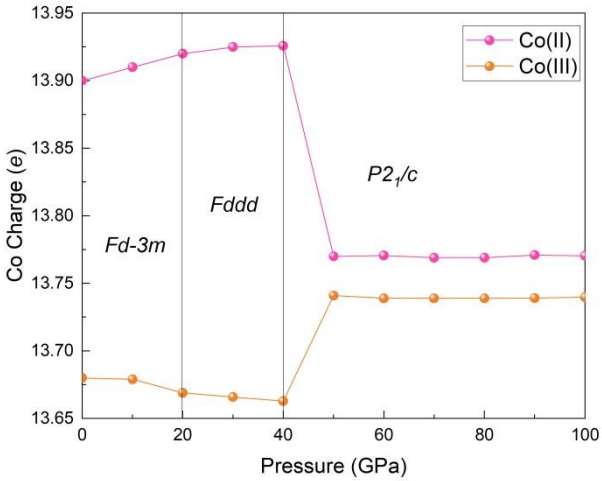


Fig. 8 Pressure-dependent Bader charges of Co^{2+} and Co^{3+} in the *Fd-3m*, *Fddd*, and $P2_1/c$ phases.

3.6 Finite-temperature free energy and phase diagram

To investigate finite-temperature effects, we calculated the free energy of each phase within the quasi-harmonic approximation based on the phonon density of states. For the *Fd-3m* and *Fddd* phases, different magnetic configurations (AFM, FM, and NM) were considered to assess the interplay between magnetic order and lattice vibrations. The Helmholtz free energy at finite temperature is expressed as: $F(V,T) = E_{sta}(V,T) + F_{zp}(V,T) + F_{vib}(V,T)$, where E_{sta} is the first-principles 0 K total energy of a static lattice at volume V, F_{zp} is the zero-point motion energy of the lattice, F_{vib} is the vibrational energy of the lattice ions. The Gibbs free energy is then obtained via $G(P,T) = F(V,T) + PV$.

Figure 9(a) presents the relative Gibbs free energies at 300 K using AFM-*Fd-3m* as a reference. The results indicate that AFM *Fd-3m* becomes thermodynamically unstable with increasing pressure and transforms into FM *Fddd* at approximately 27 GPa, followed by a transition to the $P2_1/c$ phase near 48 GPa. These transition pressures are consistent with the 0 K enthalpy results, indicating that vibrational contributions of 300 K do not qualitatively alter the phase sequence, but only introduce minor quantitative shifts. Notably, the experimentally reported *C2/m* phase remains substantially higher in Gibbs free energy over the investigated pressure range.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Combined with its dynamical instability, this result further supports that *C2/m* does not correspond to an equilibrium high-pressure phase. Instead, it is more plausibly interpreted as a metastable state stabilized by kinetic effects during decompression, rather than a thermodynamically favored configuration within the DFT+U framework.

Based on the calculated free energies, we constructed the pressure-temperature (P–T) phase diagram of Co_3O_4, as shown in Fig. 9(b). The pressure-induced structural and magnetic transitions can be understood as arising from the competition between lattice compression and electronic energy stabilization. With increasing pressure, the $Co^{2+}O_4$ tetrahedra contract more strongly than the $Co^{3+}O_6$ octahedra, producing a compressibility mismatch within the interconnected polyhedral network and driving the symmetry-lowering distortion from the cubic *Fd-3m* phase to the orthorhombic *Fddd* structure. Simultaneously, enhanced Co–O hybridization broadens the Co-3*d* bandwidth, thereby reducing the exchange stabilization energy of the localized AFM state and favoring the FM configuration. Upon further compression in the $P2_1/c$ phase, continued bandwidth broadening together with intersite charge transfer between Co^{2+} and Co^{3+} suppresses exchange splitting, ultimately resulting in magnetic collapse.

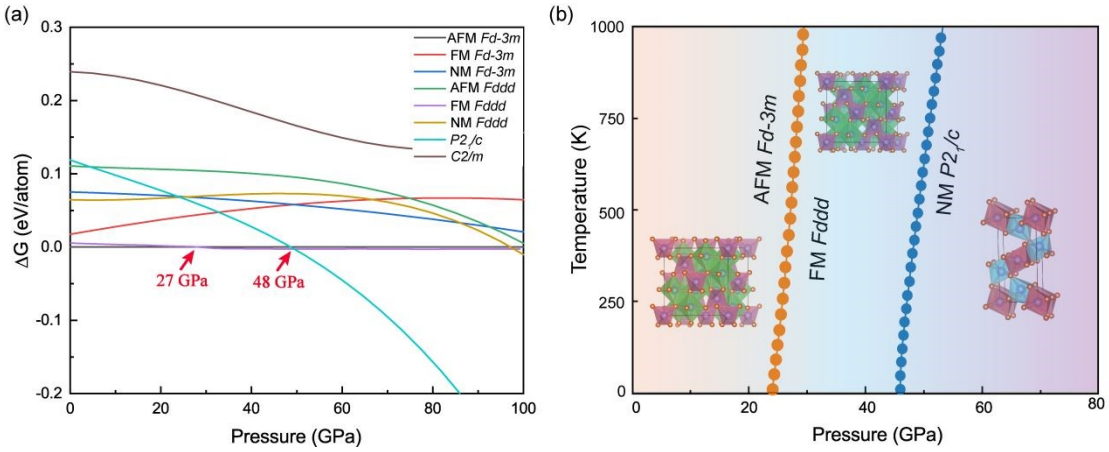


Fig. 9 (a) Calculated Gibbs free energy differences (ΔG) of the competing phases as a function of pressure at 300 K. The AFM *Fd-3m* phase is taken as the reference; (b) pressure-temperature phase diagram of Co_3O_4 constructed from the calculated Gibbs free energies.

4. Conclusions

In summary, this work systematically investigates the high-pressure phase

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

transitions, structural stability, magnetic evolution and electronic properties of Co_3O_4 over the pressure range of 0-100 GPa using DFT/DFT+U combined with particle swarm optimization structural searches. Our results demonstrate that the cubic *Fd-3m* phase first transforms into an orthorhombic *Fddd* structure through a symmetry-lowering distortion and subsequently undergoes a first-order transition to a monoclinic $P2_1/c$ phase at higher pressures. These structural transitions are accompanied by magnetic evolution from antiferromagnetic to ferromagnetic followed by magnetic collapse and the emergence of a nonmagnetic $P2_1/c$ phase. The coordination change of Co^{2+} ions and enhanced Co–O hybridization under compression play important roles in driving both structural and magnetic transitions. This work clarifies the coupled relationship between crystal structure and magnetism in Co_3O_4 under pressure and provides insight into pressure-induced phase transitions in spinel-type transition-metal oxides.

Associated Content

Notes

The authors declare no competing financial interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Author Information

Yaping Xie−Key Laboratory of Intelligent Optoelectronic Devices and Chips of Jiangsu Higher Education Institutions, School of Physical Science and Technology, Suzhou University of Science and Technology, Suzhou, 215009, China; School of Chemical Engineering, North China University of Science and Technology, Tangshan, 063210, China; Email: 2276970219@qq.com

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Ming Li—School of Chemical Engineering, North China University of Science and Technology, Tangshan, 063210, China; Email: vrchem@hotmail.com.

Cong Li—Key Laboratory of Intelligent Optoelectronic Devices and Chips of Jiangsu Higher Education Institutions, School of Physical Science and Technology, Suzhou University of Science and Technology, Suzhou, 215009, China; orcid.org/0000-0001-6571-4213; Email: cong.li@hpstar.ac.cn.

Acknowledgment

The authors acknowledge funding support from the National Natural Science Foundation of China (Grants No. 12504022), the Natural Science Foundation of the Jiangsu Higher Education Institutions of China (25KJB140018). The calculations in this work were performed at the TianheII supercomputer in Guangzhou.

References

1 Y. Liang, Y. Li, H. Wang, J. Zhou, J. Wang, T. Regier and H. Dai, *Nat. Mater.*, 2011, **10**, 780–786.

2 C. W. Tung, Y. Y. Hsu and Y. P. Shen, *Nat. Commun.*, 2015, **6**, 8106.

3 X. W. Lou, D. Deng and J. Y. Lee, *J. Mater. Chem*., 2008, **18**, 4397–4401.

4 P. Poizot, S. Laruelle and S. Grugeon, *Nature*, 2000, **407**, 496–499.

5 D. Bergman, J. Alicea and E. Gull, *Nat. Phys*., 2007, **3**, 487–491.

6 W. Chen, C. Chen and L. Guo, *J. Appl. Phys*. 2010, **108**, 073702.

7. P. G. Choi, T. Fuchigami, K. Kakimoto, *et al*., *ACS Sensors*, 2020, **5**, 1665–1673.

8. Y. Li, B. Tan and Y. Wu, *J. Am. Chem. Soc*., 2006, **128**, 14258–14259.

9. C. N. P. da Fonseca, M.-A. De Paoli and A. Gorenstein, *Adv. Mater.*, 1991, **3**, 553.

10 Y. Lou, J. Liang, Y. Peng and J. Chen, *Phys. Chem. Chem. Phys*., 2015, **17**, 8885.

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online DOI: 10.1039/D5CP05056A

11 D. Liu, X. Wang, X. Wang, W. Tian, Y. Bando and D. Golberg, *Sci. Rep.*, 2013, **3**, 2543.

12 N. Zhang, J. Shi, F. Niu, J. Wang and L. Guo, *Phys. Chem. Chem. Phys.*, 2015, **17**, 21397.

13 X. Xie, Y. Li, Z.-Q. Liu, M. Haruta and W. Shen, *Nature*, 2009, **458**, 746.

14 Z. Chen, C. X. Kronawitter and B. E. Koel, *Phys. Chem. Chem. Phys.*, 2015, **17**, 29387.

15 A. Dewaele and L. Nataf, *Phys. Rev. B*, 2022, **106**, 014104.

16 C. Wang, C. Li, J. Han, *et al. Phys. Chem. Chem. Phys.*, 2017, **19**, 22061–22068.

17 C. Li, C. Wang C, J. Han, *et al.*, *J. Mater. Sci.*, 2018, **53**, 7475–7485.

18 X. Zhang, W. Li, J. Feng, *et al.*, *Appl. Phys. Lett.*, 2025, **127**, 222601.

19 C. Li, *J Chem. Phys.*, 2024, **160**, 144302.

20 C. Li, R. T. Howie, H. Dong, *et al.*, *Nano Lett.*, 2025, **25**, 2096–2102.

21 L. Bai, M. Pravica and Y. Zhao, *J. Phys.: Condens. Matter.*, 2012, **24**, 435401.

22 N. O. Golosova, D. P. Kozlenko, D. Nicheva, *et al.*, *J. Magn. Magn. Mater.*, 2020, **508**, 166874.

23 S. Hirai and W. L. Mao, *Appl. Phys. Lett.*, 2013, **102**, 041910.

24 T. Kaewmaraya, W. Luo, X. Yang, P. Panigrahi and R. Ahuja, *Phys. Chem. Chem. Phys.*, 2015, **17**, 19957–19961.

25 O. M. Sousa, J. S. Lima, A. F. Lima, *et al.*, *J. Magn. Magn. Mater.*, 2019, **484**, 21–30.

26 F. Cova, M. V Blanco and M. Hanfland, *Phys. Rev. B*, 2019, **100**, 054111.

27 E. Mijit, K. Chen and J. E. F. S. Rodrigues, *Phys. Rev. B*, 2021, **103**, 024105.

28 Y. Wang, J. Lv, L. Zhu and Y. Ma, *Comput. Phys. Commun.*, 2012, **183**, 2063–2070.

29 Y. Wang, J. Lv, L. Zhu and Y. Ma, *Phys. Rev. B*, 2010, **82**, 094116.

30 G. Kresse and J. Furthmüller, *Phys. Rev. B*, 1996, **54**, 11169.

31 J. Hafner, *J. Comput. Chem.*, 2008, **29**, 2044–2078.

32 C. Li, W, Li, X. Zhang, L. Du and H. Sheng, *Phys. Chem. Chem. Phys.*, 2022, **24**, 12260–12266.

33 Y. Qin, X. Zhang, S. Zhou, *et al*., *J Chem. Phys.*, 2025, **162**, 014501.

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

34 C. Li, W. Yang and H. W. Sheng, *Phys. Rev. Mater.*, 2022, **6**, 033601.
35 W. Li, J. Feng, X. Zhang, *et al.*, *J. Am. Chem. Soc*., 2021, **143**, 20343–20355.
36 J. P. Perdew, K. Burke and M. Ernzerhof, *Phys. Rev. Lett.*, 1996, **77**, 3865.
37 V. I. Anisimov, J. Zaanen and O. K. Andersen, *Phys. Rev. B*, 1991, **44**, 943.
38 P. E. Blöchl, *Phys. Rev. B*, 1994, **50**, 17953.
39 H. J. Monkhorst and J. D. Pack, *Phys. Rev. B*, 1976, **13**, 5188.
40 A. Togo, *J. Phys. Soc. Japn.*, 2023, **92**, 012001.
41 A. Togo and I. Tanaka, *Scr. Mater*., 2015, **108**, 1–5.
42 R. F. Bader, *Acc. Che. Res.*, 1985, **18**, 9.
43 G. Henkelman, A. Arnaldsson and H. Jónsson, *Comput. Mater. Sci*., 2006, **36**, 354.
44 A. T. Lee, H. Park and S. Ismail-Beigi, *Phys. Rev. B*, 2021, **103**, 125105.
45 J. Chen and A. Selloni, *Phys. Rev. B*, 2012, **85**, 085306.
46 W. Hu, X. M. Cao and P. Hu, *J. Phys. Chem. C*, 2018, **122**, 19593–19602.
47 W. L. Roth, *J. Phys. Chem. Solids*, 1964, **25**, 1–10.
48 C. S. Cheng, M. Serizawa, H. Sakata, *et al*., *Mater. Chem. Phys.*, 1998, **53**, 225–230.
49 V. R. Shinde, S. B. Mahadik, T. P. Gujar, *et al*., *Appl. Surf. Sci*., 2006, **252**, 7487–7492.
50 K. J. Kim and Y. R. Park, *Solid State. Commun*., 2003, **127**, 25–28.
51 A. Cadi-Essadek, A. Roldan, D. Santos-Carballal, *et al*., *S. Afr. J. Chem*., 2021, **74**, 8–16.
52 N. H. M. Zaki, M. Mustaffa, M. F. M. Taib, *et al.*, *Int. J. Eng. Technol.*, 2018, **7**, 121–125.
53 S. Selcuk and A. Selloni, *J. Phys. Chem. C*, 2015, **119**, 9973–9979.
54 D. P. Rai, L. A. Fomin, I. V. Malikov, *et al*., *RSC Adv.*, 2020, **10**, 44633–44640.
55 Y. Peng, H. Hajiyani and R. Pentcheva, *ACS Catal.*, 2021, **11**, 5601–5613.
56 D. Scheerlinck and S. Hautecler, *Phys. Status Solidi B*, 1976, **73**, 223–228.
57 A. Dittmer, T. L. da Costa Gouveia, K. Sivalingam, *et al.*, *Phys. Chem. Chem. Phys.*, 2025, **27**, 17225-17244.

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript

View Article Online
DOI: 10.1039/D5CP05056A

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Published on 10 March 2026. Downloaded by Soochow University China on 3/10/2026 9:06:42 AM.

Physical Chemistry Chemical Physics Accepted Manuscript